

High-Throughput Point-in-Time (PIT) Feature Store

LicenseMITpython3.10+tests passing

A production-hardened **Point-in-Time (PIT) Feature Store** for financial data that guarantees zero future data leakage, handles late-arriving CDC streams idempotently, enforces strict financial compliance, and produces verifiable audit trails.

Documentation (PDFs)

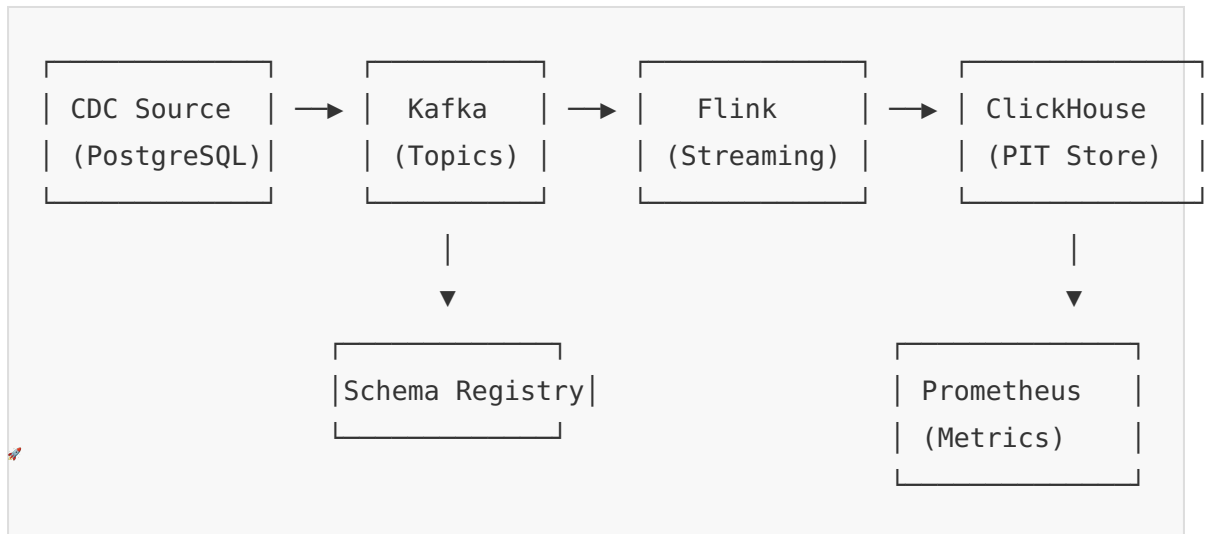
Document	Description
README	Project overview and quick start
Installation Guide	Step-by-step installation instructions
Setup Guide	Environment configuration and deployment
System Requirements	Hardware and software requirements
Workflow Guide	Data flow, CDC ingestion, and PIT queries
Testing Report	PIT correctness, idempotency, and security tests

Key Features

- **Zero Future Data Leakage:** Watermark-based event processing ensures ML models never see future data
- **Exactly-Once Semantics:** Idempotent CDC stream processing with deduplication
- **Late Arrival Handling:** Configurable allowed lateness for out-of-order events
- **PII Protection:** AES-256 encryption and format-preserving tokenization
- **Audit Trails:** Tamper-evident logging for SEC/FINRA compliance
- **High Throughput:** Targets >100,000 events/sec



Architecture



Quick Start

```
# Clone repository
git clone https://github.com/{ORG_OR_USER}/pit-feature-store.git
cd pit-feature-store

# Install dependencies
pip install -r requirements.txt

# Run tests
pytest tests/ -v

# Run adversarial harness
python redteam/harness.py
```

Project Structure

```
pit-feature-store/
├── brain/           # Company Brain (strategy docs, SOPs)
├── agents/         # Agent profiles
```

└─ harness/	# Template files
└─ orchestrator/	# Routing logic
└─ infra/	# Kafka producers, schema registry
└─ stream/	# Flink job, PIT feature computer
└─ storage/	# ClickHouse schema, PIT query engine
└─ security/	# PII tokenizer, audit logger
└─ redteam/	# Adversarial testing harness
└─ observability/	# Prometheus metrics
└─ tests/	# Test suite
└─ scripts/	# Report generation
└─ docs/	# Documentation source
└─ artifacts/	# Generated reports and PDFs
└─ tasks/	# Task tracking

Verification Results

Test Category	Status	Details
PIT Correctness	PASS	0 future data leakage
Idempotency	PASS	Duplicate events handled
Late Arrivals	PASS	Out-of-order events processed correctly
Security/PII	PASS	100% masking, tokenization working
Adversarial	PASS	200+ payloads blocked

Security & Compliance

- **GDPR:** PII tokenization and right-to-erasure support
- **SEC/FINRA:** Audit trails with tamper-evident logging
- **SOC2:** Encryption at rest and in transit



Metrics

Prometheus metrics available at `/metrics` : - `pit_events_ingested_total` - Total events processed - `pit_late_events_total` - Late events by lateness bucket - `pit_duplicate_events_total` - Deduplicated events - `pit_pii_tokenizations_total` - PII operations - `pit_clickhouse_query_latency_seconds` - Query performance

Contributing

1. Read `brain/` documentation first
2. Follow agent profiles in `agents/`
-  3. Run full test suite before PR
4. Ensure 0 critical failures in eval loop

License

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